

Tutorial 8: System of First-order Linear DEs (Part 1)

1. Rewrite the following linear DE as a system of first-order DE:

$$y''' + 3xy'' + y' \sin x + ye^x = 2x$$

and further formulate the result in a matrix.

Solution: Introduce three unknowns

$$X_1 = y, \quad X_2 = y', \quad X_3 = y''.$$

Then we can rewrite the given ODE as

$$X_3' + 3xX_3 + X_2 \sin x + X_1 e^x = 2x$$

We formulate them as the standard form

$$\begin{cases} X_1' = X_2, \\ X_2' = X_3, \\ X_3' = -e^x X_1 - \sin x X_2 - 3x X_3 + 2x. \end{cases}$$

Formulate it as the matrix form: $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}$, i.e.,

$$\frac{d}{dx} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -e^x & -\sin x & -3x \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2x \end{pmatrix}.$$

2. i) Write down the Abel's formula for the 3-by-3 system of ODEs: $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$, where $\mathbf{A} = (a_{ij})_{1 \leq i, j \leq 3}$ is a 3-by-3 square matrix.
- ii) Derive the Abel's formula for the third-order equation

$$y''' + p_1 y'' + p_2 y' + p_3 y = 0,$$

from i) by writing it as a system of 1st-order DEs.

Solution: i) The Abel's formula for the system is

$$W(t) = c \exp\left(\int \operatorname{tr}(\mathbf{A}) dt\right) = c \exp\left(\int (a_{11} + a_{22} + a_{33}) dt\right),$$

where $W(t)$ is the Wronskian of any three solutions of the system.

ii) Similar to the first question, we can rewrite it as a system of first-order ODEs by introducing

$$X_1 = y, \quad X_2 = y', \quad X_3 = y'',$$

that is,

$$\frac{d}{dx} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -p_3 & -p_2 & -p_1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}.$$

The trace of the coefficient matrix $\mathbf{A}$: $\operatorname{tr}(\mathbf{A}) = -p_1$, so by i),

$$W(x) = c \exp\left(-\int p(x) dx\right).$$

Note: $W(x) = W[y_1, y_2, y_3](x)$ where y_1, y_2, y_3 are three solutions of the third-order DE. Correspondingly, we have the three solutions of the system:

$$\mathbf{X}_i = \begin{pmatrix} y_i \\ y'_i \\ y''_i \end{pmatrix}, \quad i = 1, 2, 3.$$

Therefore,

$$W(x) = W[y_1, y_2, y_3](x) = W[\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3](x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y'_1 & y'_2 & y'_3 \\ y''_1 & y''_2 & y''_3 \end{vmatrix}.$$

This verifies the Abel's formula.

3. Solve the system

$$x'_1 = x_1 + 2x_2, \quad x'_2 = 2x_1 - 2x_2,$$

by eliminating x_2 and converting the system to a second-order equation.

Solution: Differentiate the first equation and use the second equation:

$$\begin{aligned}x_1'' &= x_1' + 2x_2' = x_1' + 2(2x_1 - 2x_2) = x_1' + 4x_1 - 4x_2 \\ &= x_1' + 4x_1 - 2(x_1' - x_1) = -x_1' + 6x_1.\end{aligned}$$

This leads to the second-order ODE:

$$x_1'' + x_1' - 6x_1 = 0 \quad \Rightarrow \quad \text{CE : } r^2 + r - 6 = (r + 3)(r - 2) = 0.$$

Thus, its general solution is

$$x_1(t) = c_1 e^{-3t} + c_2 e^{2t}.$$

Then by the first equation

$$x_2 = \frac{1}{2}(x_1' - x_1) = -2c_1 e^{-3t} + \frac{c_2}{2} e^{2t}.$$

Finally, we formulate the general solution as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix} e^{2t}.$$

4. Suppose that the Wronskian of n solutions to a linear system $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$, $t \in I = (-3, 3)$, is $W(t)$.
- (i) If $W(t) = \frac{1}{2}$, then what property the matrix $\mathbf{A}(t)$ has.
 - (ii) If $W(0) = 1$, is it possible that there is one point $t_0 \in I$, such that $W(t_0) < 0$? (Explain why).

Solution: (i) Recall the Abel's formula:

$$W'(t) = \text{tr}(\mathbf{A})W(t) \quad \Leftrightarrow \quad W(t) = c \exp\left(\int \text{tr}(\mathbf{A})dt\right).$$

As $W(t) = \frac{1}{2}$, we find from the first formula that

$$(1/2)' = 0 = \text{tr}(\mathbf{A})\frac{1}{2} \quad \Rightarrow \quad \text{tr}(\mathbf{A}) = 0.$$

- (ii) We can rewrite the Abel's formula in the definite integral form

$$W(t) = c \exp\left(\int \operatorname{tr}(\mathbf{A}) dt\right) = \tilde{c} \exp\left(\int_0^t \operatorname{tr}(\mathbf{A}) dt\right),$$

as the two integrals only differ from some constant, i.e.,

$$\int \operatorname{tr}(\mathbf{A}) dt = \int_0^t \operatorname{tr}(\mathbf{A}) dt + C.$$

As $W(0) = 1$, we find $\tilde{c} = 1$, so

$$W(t) = \exp\left(\int_0^t \operatorname{tr}(\mathbf{A}) dt\right) > 0, \quad \forall t \in (-3, 3).$$

Therefore, we conclude that it is impossible to find a point $t_0 \in I$, such that $W(t_0) < 0$.

5. Find the general solution to

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x}.$$

Solution: We denote the coefficient matrix by $\mathbf{A}$, and first find its eigenvalues by solving

$$\begin{aligned} 0 = \det(\mathbf{A} - \lambda \mathbf{I}_3) &= \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -\lambda \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 1 \\ 1 & -\lambda \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & -\lambda \\ 1 & 1 \end{vmatrix} \\ &= -\lambda(\lambda^2 - 1) + (\lambda + 1) + (1 + \lambda) = (\lambda + 1)(2 - \lambda(\lambda - 1)) \\ &= -(\lambda + 1)^2(\lambda - 2). \end{aligned}$$

Therefore, the eigenvalues are $\lambda_1 = -1$ (multiplicity=2) and $\lambda_2 = 2$.

We next look for the corresponding eigenvectors. For $\lambda_1 = -1$, we solve the linear system:

$$(\mathbf{A} - \lambda_1 \mathbf{I}_3) \mathbf{v} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

that is

$$v_1 + v_2 + v_3 = 0.$$

It is one equation with two unknowns, so we have two free variables which we set $v_2 = r, v_3 = s$. Then we have the solution:

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} -r - s \\ r \\ s \end{pmatrix} = r \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

In other words, the eigen-space is spanned by the two LI eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Then we obtain two solutions from the eigen-pairs:

$$\mathbf{x}_1(t) = e^{-t}\mathbf{v}_1 = e^{-t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2(t) = e^{-t}\mathbf{v}_2 = e^{-t} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

We continue to find the corresponding eigenvector of $\lambda_2 = 2$, by solving

$$(\mathbf{A} - \lambda_2 \mathbf{I}_3)\mathbf{v} = \mathbf{0} \Rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

As in linear algebra, we reduce the coefficient matrix into the row echelon form using row operations:

$$\begin{aligned} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} : (R_1/2 + R_2 \rightarrow R_2) &\Rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 3/2 & -3/2 \end{pmatrix} \\ : (R_2 + R_3 \rightarrow R_3) \& R_3 \times 2/3 \rightarrow R_3 \Rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Therefore, the rank is 2 so it has one free variable, we choose $v_3 = r$, then $v_1 = v_2 = r$, and eigenvectors are

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = r \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \forall r \neq 0.$$

Therefore, we obtain one solution from this pair:

$$\mathbf{x}_3(t) = e^{2t}\mathbf{v} = e^{2t} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Finally, we get the general solution

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + c_3 e^{2t} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

6. Find the general solution to

$$\mathbf{x}' = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} \mathbf{x}.$$

Solution: We denote the coefficient matrix by $\mathbf{A}$, and first find its eigenvalues by solving

$$\begin{aligned} 0 = \det(\mathbf{A} - \lambda \mathbf{I}_3) &= \begin{vmatrix} -3 - \lambda & 0 & 2 \\ 1 & -1 - \lambda & 0 \\ -2 & -1 & -\lambda \end{vmatrix} \\ &= (-3 - \lambda) \begin{vmatrix} -1 - \lambda & 0 \\ -1 & -\lambda \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & -1 - \lambda \\ -2 & -1 \end{vmatrix} \\ &= -\lambda(\lambda + 1)(\lambda + 3) - 6 - 4\lambda \\ &= -(\lambda^3 + 4\lambda^2 + 7\lambda + 6) = -(\lambda + 2)(\lambda^2 + 2\lambda + 3), \end{aligned}$$

where the factorization is based on the inspection: $\lambda = -2$ is a root. Therefore, the eigenvalues are

$$\lambda_{1,2} = -1 \pm \sqrt{2}i, \quad \lambda_3 = -2.$$

We next look for the corresponding eigenvectors. For $\lambda_1 = -1 + \sqrt{2}i$, we solve the linear system:

$$(\mathbf{A} - \lambda_1 \mathbf{I}_3) \mathbf{v} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} -2 - \sqrt{2}i & 0 & 2 \\ 1 & -\sqrt{2}i & 0 \\ -2 & -1 & 1 - \sqrt{2}i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

We now use the row operations to reduce the matrix: i) $R_1 * (-2 + \sqrt{2}i)/6 \rightarrow R_1$; ii)

$R_2 - R_1 \rightarrow R_1$ & $R_3 + 2 * R_1 \rightarrow R_3$; iii) $R_2/(-\sqrt{2}i)$ & $R_3/(-1)$; iv) $R_3 - R_2 \rightarrow R_3$:

$$\begin{aligned} & \begin{pmatrix} -2 - \sqrt{2}i & 0 & 2 \\ 1 & -\sqrt{2}i & 0 \\ -2 & -1 & 1 - \sqrt{2}i \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & \frac{1}{3}(-2 + \sqrt{2}i) \\ 1 & -\sqrt{2}i & 0 \\ -2 & -1 & 1 - \sqrt{2}i \end{pmatrix} \\ & \rightarrow \begin{pmatrix} 1 & 0 & \frac{1}{3}(-2 + \sqrt{2}i) \\ 0 & -\sqrt{2}i & \frac{1}{3}(2 - \sqrt{2}i) \\ 0 & -1 & -\frac{1}{3}(1 + \sqrt{2}i) \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & \frac{1}{3}(-2 + \sqrt{2}i) \\ 0 & 1 & \frac{1}{3}(1 + \sqrt{2}i) \\ 0 & 1 & \frac{1}{3}(1 + \sqrt{2}i) \end{pmatrix} \\ & \rightarrow \begin{pmatrix} 1 & 0 & \frac{1}{3}(-2 + \sqrt{2}i) \\ 0 & 1 & \frac{1}{3}(1 + \sqrt{2}i) \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

Choose $v_3 = r$ to be the free variable. We find the eigenvectors

$$\mathbf{v} = r \begin{pmatrix} -\frac{1}{3}(-2 + \sqrt{2}i) \\ -\frac{1}{3}(1 + \sqrt{2}i) \\ 1 \end{pmatrix}, \quad \forall r \neq 0.$$

We take $r = -3$ and form the solution. Then we use Euler's formula and calculate its real and imaginary parts:

$$\begin{aligned} e^{(-1+\sqrt{2}i)t} \begin{pmatrix} -2 + \sqrt{2}i \\ 1 + \sqrt{2}i \\ -3 \end{pmatrix} &= e^{-t} (\cos(\sqrt{2}t) + i \sin(\sqrt{2}t)) \begin{pmatrix} -2 + \sqrt{2}i \\ 1 + \sqrt{2}i \\ -3 \end{pmatrix} \\ &= e^{-t} \begin{pmatrix} -2 \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) + i(\sqrt{2} \cos(\sqrt{2}t) - \sin(\sqrt{2}t)) \\ \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) + i(\sqrt{2} \cos(\sqrt{2}t) + \sin(\sqrt{2}t)) \\ -3 \cos(\sqrt{2}t) - 3i \sin(\sqrt{2}t) \end{pmatrix} \\ &= e^{-t} \begin{pmatrix} -2 \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) \\ \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) \\ -3 \cos(\sqrt{2}t) \end{pmatrix} + i e^{-t} \begin{pmatrix} \sqrt{2} \cos(\sqrt{2}t) - \sin(\sqrt{2}t) \\ \sqrt{2} \cos(\sqrt{2}t) + \sin(\sqrt{2}t) \\ -3 \sin(\sqrt{2}t) \end{pmatrix} \end{aligned}$$

We then obtain two LI solutions from the real and imaginary parts as follows:

$$\mathbf{x}_1 = e^{-t} \begin{pmatrix} -2 \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) \\ \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) \\ -3 \cos(\sqrt{2}t) \end{pmatrix}, \quad \mathbf{x}_2 = e^{-t} \begin{pmatrix} \sqrt{2} \cos(\sqrt{2}t) - \sin(\sqrt{2}t) \\ \sqrt{2} \cos(\sqrt{2}t) + \sin(\sqrt{2}t) \\ -3 \sin(\sqrt{2}t) \end{pmatrix}$$

Next, we seek the eigenvector for $\lambda_3 = -2$ by solving the system

$$(\mathbf{A} - \lambda_3 \mathbf{I}_3) \mathbf{v} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} -1 & 0 & 2 \\ 1 & 1 & 0 \\ -2 & -1 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

which can be reduced to

$$\begin{pmatrix} -1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We set $v_3 = s$, and obtain the eigenvectors

$$\mathbf{v} = s \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, \quad \forall s \neq 0.$$

We obtain the solution:

$$\mathbf{x}_3 = e^{-2t} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}.$$

Therefore, the general solution is

$$\begin{aligned} \mathbf{x}(t) = & c_1 e^{-t} \begin{pmatrix} -2 \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) \\ \cos(\sqrt{2}t) - \sqrt{2} \sin(\sqrt{2}t) \\ -3 \cos(\sqrt{2}t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \sqrt{2} \cos(\sqrt{2}t) - 2 \sin(\sqrt{2}t) \\ \sqrt{2} \cos(\sqrt{2}t) + \sin(\sqrt{2}t) \\ -3 \sin(\sqrt{2}t) \end{pmatrix} \\ & + c_3 e^{-2t} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}. \end{aligned}$$

7. Consider the system of linear equations

$$\begin{cases} x_1'' - 2x_1' - x_2' + 2x_2 = 0, \\ x_1' - 2x_1 + x_2' = 0, \end{cases}$$

with the initial condition

$$x_1(0) = 3, \quad x_1'(0) = 2, \quad x_2(0) = 0.$$

Introduce $x_3 = x_1'$ and rewrite it as a system of first-order equations in the matrix form $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ with $\mathbf{x} = (x_1, x_2, x_3)^T$. Find its general solution.

Solution: As $x_1' = x_3$, we can reduce the system as

$$\begin{cases} x_3' - 2x_3 - x_2' + 2x_2 = 0, \\ x_3 - 2x_1 + x_2' = 0, \end{cases}$$

From the second equation, we have $x'_2 = 2x_1 - x_3$, so we obtain from the first equation that

$$x'_3 = 2x_3 + x'_2 - 2x_2 = 2x_3 + (2x_1 - x_3) - 2x_2 = 2x_1 - 2x_2 + x_3.$$

In summary, we have

$$\begin{cases} x'_1 = x_3, \\ x'_2 = 2x_1 - x_3, \\ x'_3 = 2x_1 - 2x_2 + x_3 \end{cases} \Rightarrow \mathbf{x}'(t) = \begin{pmatrix} 0 & 0 & 1 \\ 2 & 0 & -1 \\ 2 & -2 & 1 \end{pmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}.$$

We denote the coefficient matrix by $\mathbf{A}$. We can find the eigenvalues $\lambda_1 = 1, \lambda_2 = -2$ and $\lambda_3 = 2$. Then we can find the corresponding eigenvectors.

(i) For $\lambda_1 = 1$, we solve

$$\begin{pmatrix} -1 & 0 & 1 \\ 2 & -1 & -1 \\ 2 & -2 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and reduce it to

$$\begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

So the eigenvector and solution are

$$\mathbf{v} = r \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{x}_1(t) = e^t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

(ii) For $\lambda_1 = -2$, we solve

$$\begin{pmatrix} 2 & 0 & 1 \\ 2 & 2 & -1 \\ 2 & -2 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and reduce it to

$$\begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

So the eigenvector and solution are

$$\mathbf{v} = r \begin{pmatrix} -\frac{1}{2} \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{x}_2(t) = e^{-2t} \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}.$$

(iii) For $\lambda_3 = 2$, we solve

$$\begin{pmatrix} -2 & 0 & 1 \\ 2 & -2 & -1 \\ 2 & -2 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and reduce it to

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

So the eigenvector and solution are

$$\mathbf{v} = r \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{x}_3(t) = e^{2t} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

Therefore, the general solution of the system becomes

$$\mathbf{x}(t) = c_1 e^t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + c_3 e^{2t} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

Now, we determine the constants by using the initial condition

$$\mathbf{x}(0) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}.$$

Solving this system leads to

$$c_1 = \frac{4}{3}, \quad c_2 = -\frac{2}{3}, \quad c_3 = 1.$$

Thus, the solution is

$$\mathbf{x}(t) = \frac{4}{3} e^t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3} e^{-2t} \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + e^{2t} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$